Building with Blueprints

THE QUIZ MASTER | SESSION 1

Coding is like magic—you build your own world with simple blueprints! [cite: 279] Today, we are moving beyond simple lines of code to become master architects of our own Quiz Game. [cite: 142]

The Challenge of Manual Quizzing

Imagine trying to manage 100 questions one by one. [cite: 154] Without a plan, your code becomes messy and hard to track. [cite: 168, 169] It's easy to lose track of which questions you've already asked! [cite: 167]

The Question: What is the national fruit of India? [cite: 152, 264]
The Answer: Mango [cite: 153, 265]

Now imagine writing this code 100 times. There must be a smarter way! [cite: 170]

Class vs. Object: The Blueprint

In Python, we use something called a **Class** to solve our messy code problem. [cite: 200, 276]

THE RUBBER STAMP ANALOGY

Think of a **Class** like a rubber stamp design. You create the design once. [cite: 178, 195]

Every time you press that stamp on paper, you get a real ink mark—this is the **Object**! [cite: 179, 197]

One Stamp Design = Many Ink Marks! [cite: 180]

CLASS = BLUEPRINT: The idea or design template. [cite: 194]

OBJECT = REAL THING: The actual thing created from the template. [cite: 196]

Our 'Question' Blueprint

Let's write our first blueprint in Python code. This will be the design for *every* quiz question we ever make! [cite: 201]

THE BLUEPRINT CODE

```
class Question:
    def __init__(self, text, answer):
        self.text = text
        self.answer = answer
```

WHAT DOES THIS DO?

- **Class Question:** This tells Python we are making a new blueprint. [cite: 204]
- **The init Method:** This is a special method that sets up each object's information. [cite: 205, 277]
- **Attributes:** These are the pieces of information each object holds—like the question text and its answer! [cite: 193, 206, 207]

Creating Real Questions

Now that we have our stamp (the Class), let's make some ink marks (the Objects)! [cite: 215]

CODE EXAMPLE

```
q1 = Question("Capital of India?", "New Delhi")
q2 = Question("IPL 2023 winner?", "CSK")
q3 = Question("ISRO chief?", "Dr. S. Somanath")
```

STORING OUR OBJECTS

We can put all our "real things" into a list so they are easy to manage. [cite: 221]

```
questions = [q1, q2, q3]
```

Now we have 3 questions ready to play! [cite: 223]

How the Quiz Game Works

To run our game, we need to go through our list and do three things for every question: [cite: 230]

- ✔ **Show Question Text:** Display the question on the screen so the player can read it. [cite: 231, 232]
- ✔ **Get Answer:** Wait for the player to type their answer using the `input()` function. [cite: 233, 234]
- ✔ **Check & Respond:** Compare the player's answer to the real answer and say "Correct!" or "Try again!" [cite: 235, 236, 237]

Code Tip: We use a `for` loop to go through our list of question objects one by one! [cite: 242]

The Quiz Logic

Here is how we check the answers using Python code: [cite: 241]

```
for question in questions:
    user_answer = input(question.text + " ")

    if user_answer.lower() == question.answer.lower():
        print("Correct!")
    else:
        print("Incorrect. Try next question.")
```

INDIAN GK CHALLENGE!

Can you answer these questions in your game? [cite: 262, 272]

- **Q2:** Which city is called the Silicon Valley of India? (Answer: Bengaluru) [cite: 267, 268, 269]
- **Q3:** Which Indian state is famous for the Sun Temple? (Answer: Odisha) [cite: 270, 271]

What Did We Learn Today?

- A **Class** is a blueprint for creating objects. [cite: 276]
- An **Object** is the real thing made from that blueprint! [cite: 276]
- The **__init__** method sets up each object's information. [cite: 277]
- We built a quiz game using question objects! [cite: 277]

WHAT'S NEXT? [CITE: 283]

Get ready for Session 2! We will learn how to track your points, add levels, and create a real quiz competition! [cite: 282, 283]

Coming Soon! [cite: 284]

"Coding is the power to bring your ideas to life!" [cite: 280]

Thank you for coding with us! [cite: 285]